- The culture space should be sprayed with 0.1% Nuvan and 5% Formalin, two days prior to spawning and transferring to bags to cropping rooms.
- The spawn used for mushroom should be free from contaminations. Bags should be filled with 8 kg of wet straw.
- During spawning running temperature and relative humidity should be maintained 20^0C to 30^0C, 75 to 85% respectively.
- Proper watering should be done when the growth coverings are removed. There should not be dry patches on blocks. Excess watering must be avoided.
- About 20 cm gap should be maintained in between two bags or blocks.
- Picking must be done as their caps become 10 – 12 cm by twisting.

Two kinds of mushrooms are cultivated namely **button** and **oyster**.

10.11.2 Single Cell Protein (SCP) Production

Single-cell proteins are the dried cells of microorganism, which are used as protein supplement in human foods or animal feeds. Microorganisms that can be used for the production of SCP have the capacity to synthesize proteins rapidly than higher living organisms. Microorganisms like algae, fungi, yeast and bacteria are used for this purpose. Here you will learn about the production of SCP from an algae, *Spirulina*.



Figure 10.23: Production of *Spirulina*

Small scale biomass production of *Spirulina*.

It requires an aquarium, air pump, nutrients and *Spirulina* mother culture.

- Take a 30 litre capacity aquarium and fill half of it with water.
- Check if any heavy metal concentration or fluorine or calcium carbonate in water.
- Fill the tank with water and add nutrients preferably **zarrouk medium**. (Add half of the required nutrients first and add another half later).
- To aerate the culture, fix the air pump (avoid centrifugal pump) after adding nutrients.
- Add the mother culture to the aquarium. For every 1 liter of water add 4 gm. mother culture.
- Place it in sunlight for 10-12 hrs. every day.
- After a week check the culture and add more water leave it for one more week, till the biomass becomes dark green.
- Use a very fine cloth and harvest the algae.
- Water can be reused in aquarium.
- Dry the algae for later use.

Single cell protein has a high nutritive value due to higher protein, vitamin, essential amino acids and lipid content. Hence it can form a good protein supplement. However it cannot completely replace the conventional protein sources due to their high nucleic acid content and slower in digestibility. They may result in allergic reactions.

10.11.3 Seaweed Liquid Fertilizer

Seaweed is rich in trace elements and potassium, which makes it ideal to add to compost in its raw state, to work in as a mulch, or to create a liquid fertilizer. This is easy to do. Seaweed fertilizer releases about 60 nutrients from which plants can benefit.

- Collect the seaweed that is not too stinky.
- Rinse the seaweed to remove the excess salt.
- Fill a bucket to three quarters way with

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